In fact, on closer inspection, Ramond's research, together with results found by Schwarz and his collaborator André Neveu and later insights of Ferdinando Gliozzi, Joël Scherk, and David Olive, revealed a perfect balance—a novel symmetry—between the vibrational patterns with different spins in the modified string theory. These researchers found that the new vibrational patterns arose in pairs whose spin values differed by half a unit. For every vibrational pattern with spin-½ there was an associated vibrational pattern with spin-0. For every vibrational pattern of spin-1, there was an associated vibrational pattern of spin-½, and so on. The relationship between integer and half-integer spin values was named *supersymmetry,* and with these results the *supersymmetric string theory,* or *superstring theory,* was born. Nearly a decade later, when Schwarz and Green showed that all the potential anomalies that threatened string theory canceled each other out, they were actually working in the framework of superstring theory, and so the revolution their paper ignited in 1984 is more appropriately called the first *superstring* revolution. (In what follows, we will often refer to strings and to string theory, but that's just a shorthand; we always mean superstrings and superstring theory.)

With this background, we can now state what it would mean for string theory to reach beyond broad-brush features and explain the universe in detail. It comes down to this: among the vibrational patterns that strings can execute, there must be patterns whose properties agree with those of the known particle species. The theory has vibrational patterns with spin-½, but it must have spin-½ vibrational patterns that match precisely the known matter particles, as summarized in Table 12.1. The theory has spin-1 vibrational patterns, but it must have spin-1 vibrational patterns that match *precisely* the known messenger particles, as summarized in Table 12.2. Finally, if experiments do indeed discover spin-0 particles, such as are predicted for Higgs fields, string theory must yield vibrational patterns that match *precisely* the properties of these particles as well. In short, for string theory to be viable, its vibrational patterns must yield and explain the particles of the standard model.

Here, then, is string theory's grand opportunity. If string theory is right, there *is* an explanation for the particle properties that experimenters have measured, and it's to be found in the resonant vibrational patterns that strings can execute. If the properties of these